MCU / SPI control input

ESP32-S3-DevKitC-1-N16R8 socket

3V3 distribution, pullups, and 4-wire mode

R12/R13 set BS0=0 and BS1=0 for 4-wire SPI.

Panel FPC and MF-1750 signals

MF-1750 50pin FPC

HV boost and negative gate pump

Main current path: V_B -> L1 -> SW_H/Q1/D1 -> VPH -> R5 -> PREVGH.

Panel-side caps, references, and flying capacitors

VPC positive pump

Q2/L2/D4/R3 form the VPC pump; keep this loop compact in PCB layout.

Bring-up and reserved test pads

Reference source: MF-1750/M009F-T specification page 10 reference circuit.

Verified topology: R16 fitted routes SL_CSB to DevKitC GPIO10; R15 is DNP.

Functional blocks now group and wire local nodes; net labels are reserved for cross-block interfaces.

Default bus mode: BS0=0 and BS1=0 through R12/R13 for 4-wire SPI with D/C.

Diode polarity is assigned in BOM notes; match footprint pin A/K to those notes before PCB layout.

Reserved FPC pins are routed only to isolated single test pads; pin1 NC is explicitly no-connect.

65 x 60.5 mm engineering layout; R16 independent CS	
Sheet: /	
File: mf1750_devkitc_b2_pcba_margin.kicad_sch	
Title: MF-1750 DevKitC Compact Carrier Rev B2 PCBA Edge Margin	
Size: A2	Date:
KiCad E.D.A. 10.0.3	Rev:
	Id: 1/1